

Chapter 2, Problem 3PP

Problem

A resistor absorbs an instantaneous power of $30 \cos^2 t$ mW when connected to a voltage source $v = 15 \cos t$ V. Find i and R .

Step-by-step solution

Step 1 of 2

The instantaneous power absorbed by a resistor $p = 30 \cos^2 t$ mW

The voltage across the resistor $v = 15 \cos t$ V

Write the expression for power.

$$p = vi$$

Calculate the current.

$$\begin{aligned} i &= \frac{p}{v} \\ &= \frac{30 \cos^2 t \text{ mW}}{15 \cos t \text{ V}} \\ &= 2 \cos t \text{ mA} \end{aligned}$$

Therefore, the current is $2 \cos t \text{ mA}$.

Step 2 of 2

Calculate the resistance.

$$\begin{aligned} R &= \frac{v}{i} \\ &= \frac{15 \cos t \text{ V}}{2 \cos t \text{ mA}} \\ &= 7.5 \text{ k}\Omega \end{aligned}$$

Therefore, the resistance is $7.5 \text{ k}\Omega$.